Case Study 6: Student Performance Trend Analysis

1. Problem Statement

Develop a Python program to analyze student performance across three examinations. The program should accept subject names and marks, calculate the average marks for each subject, identify whether performance has improved or declined, and identify subjects that need improvement.

2. Objectives

- To understand variables, data types, and arithmetic operators.
- To use lists and nested lists to store student marks.
- To apply loops and conditional statements.
- To calculate average marks and compare examination results.
- To identify improving, declining, and weak subjects.

3. Assumptions

1. The student has three examination scores for each subject.
2. Each examination mark must be between 0 and 100.
3. The program compares the first examination with the third examination.
4. If the third examination mark is higher, performance has improved.
5. If the third examination mark is lower, performance has declined.
6. A subject is considered weak if its average mark is below 40 or its third examination mark is below 40.

These are example assumptions for implementing the case study.

4. Algorithm

1. Start the program.
2. Ask the user to enter the number of subjects.
3. Create an empty nested list to store subject names and marks.
4. For each subject, enter marks for three examinations.

5. Validate that each mark is between 0 and 100.
6. Calculate the average marks for each subject.
7. Compare the first and third examination marks.
8. Display whether performance improved, declined, or remained unchanged.
9. Identify weak subjects using the given assumptions.
10. Display the analysis report and stop.

5. Flowchart

Start
Enter number of subjects
Enter subject names and three exam marks
Are all marks between 0 and 100?

Yes

Calculate average

No

Re-enter marks
Compare first and third exam marks
Display trends and weak subjects
Stop

6. Pseudocode

```
None
START

INPUT number_of_subjects
CREATE empty list called student_data

FOR each subject:
    INPUT subject_name

    INPUT marks for Exam 1, Exam 2, Exam 3
    VALIDATE each mark is between 0 and 100

    STORE subject name and marks in student_data
```

```
FOR each subject in student_data:
    average = total marks / 3

    IF Exam 3 > Exam 1:
        trend = "Improved"
    ELSE IF Exam 3 < Exam 1:
        trend = "Declined"
    ELSE:
        trend = "Unchanged"

    IF average < 40 OR Exam 3 < 40:
        subject is weak

    DISPLAY average, trend, and weak-subject status

STOP
```

7. Python Source Code

```
print("===== STUDENT PERFORMANCE ANALYSIS =====")


while True:
    try:
        n = int(input("Enter number of subjects: "))
        if n > 0:
            break
        print("Enter at least one subject.")
    except ValueError:
        print("Please enter a valid whole number.")
```

```
student_data = []


for i in range(n):
    print("\nSubject", i + 1)
    subject = input("Enter subject name: ")


    marks = []


    for exam in range(1, 4):
        while True:
            try:
                mark = float(
                    input(f"Enter Exam {exam} marks (0-100): ")
                )


                if 0 <= mark <= 100:
                    marks.append(mark)
                    break
                else:
                    print("Marks must be between 0 and 100.")


            except ValueError:
                print("Please enter a valid number.")
```

```
    student_data.append([subject] + marks)

print("\n===== PERFORMANCE REPORT =====")

for record in student_data:
    subject = record[0]
    exam1 = record[1]
    exam2 = record[2]
    exam3 = record[3]

    average = (exam1 + exam2 + exam3) / 3

    if exam3 > exam1:
        trend = "Improved"
    elif exam3 < exam1:
        trend = "Declined"
    else:
        trend = "Unchanged"

    if average < 40 or exam3 < 40:
        status = "Needs Improvement"
    else:
        status = "Satisfactory"
```

```
    print("\nSubject:", subject)
    print("Exam 1:", exam1)
    print("Exam 2:", exam2)
    print("Exam 3:", exam3)
    print("Average:", round(average, 2))
    print("Performance Trend:", trend)
    print("Status:", status)

print("\nAnalysis completed.")
```

```
None
```

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8. Test Cases

Test	Exam 1	Exam 2	Exam 3	Expected result
Mathematics	50	60	70	Average 60; Improved
Python	45	55	65	Average 55; Improved

Physics	60	50	35	Average 48.33; Declined; Needs Improvement
Invalid marks	110	—	—	Reject 110 and ask again

Sample Output

For this example, enter three subjects and the marks shown in the test cases above.

```
None
===== STUDENT PERFORMANCE ANALYSIS =====
Enter number of subjects: 3

Subject 1
Enter subject name: Mathematics
Enter Exam 1 marks (0-100): 50
Enter Exam 2 marks (0-100): 60
Enter Exam 3 marks (0-100): 70

Subject 2
Enter subject name: Python
Enter Exam 1 marks (0-100): 45
Enter Exam 2 marks (0-100): 55
Enter Exam 3 marks (0-100): 65

Subject 3
Enter subject name: Physics
Enter Exam 1 marks (0-100): 60
Enter Exam 2 marks (0-100): 50
Enter Exam 3 marks (0-100): 35

===== PERFORMANCE REPORT =====

Subject: Mathematics
Exam 1: 50.0
```

```
Exam 2: 60.0
Exam 3: 70.0
Average: 60.0
Performance Trend: Improved
Status: Satisfactory

Subject: Python
Exam 1: 45.0
Exam 2: 55.0
Exam 3: 65.0
Average: 55.0
Performance Trend: Improved
Status: Satisfactory

Subject: Physics
Exam 1: 60.0
Exam 2: 50.0
Exam 3: 35.0
Average: 48.33
Performance Trend: Declined
Status: Needs Improvement

Analysis completed.
```

9. Output Screenshots

```
Enter "help" below or click "Help" above for more information.
>
====== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/6.student pe
===== STUDENT PERFORMANCE ANALYSIS =====
Enter number of subjects: 3

Subject 1
Enter subject name: Mathematics
Enter Exam 1 marks (0-100): 50
Enter Exam 2 marks (0-100): 60
Enter Exam 3 marks (0-100): 70

Subject 2
Enter subject name:
========== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/6.studer
===== STUDENT PERFORMANCE ANALYSIS =====
Enter number of subjects: 2

Subject 1
Enter subject name: Python
Enter Exam 1 marks (0-100): 45
Enter Exam 2 marks (0-100): 55
Enter Exam 3 marks (0-100): 65

Subject 2
Enter subject name: Mathematics
Enter Exam 1 marks (0-100): 50
Enter Exam 2 marks (0-100): 60
Enter Exam 3 marks (0-100): 70

===== PERFORMANCE REPORT =====

Subject: Python
Exam 1: 45.0
Exam 2: 55.0
Exam 3: 65.0
Average: 55.0
Performance Trend: Improved
Status: Satisfactory

Subject: Mathematics
Exam 1: 50.0
Exam 2: 60.0
Exam 3: 70.0
Average: 60.0
Performance Trend: Improved
Status: Satisfactory

Analysis completed
```

```
print("===== STUDENT PERFORMANCE ANALYSIS =====")

while True:
    try:
        n = int(input("Enter number of subjects: "))
        if n > 0:
            break
        print("Enter at least one subject.")
    except ValueError:
        print("Please enter a valid whole number.")

student_data = []

for i in range(n):
    print("\nSubject", i + 1)
    subject = input("Enter subject name: ")

    marks = []

    for exam in range(1, 4):
        while True:
            try:
                mark = float(
                    input(f"Enter Exam {exam} marks (0-100): ")
                )

                if 0 <= mark <= 100:
                    marks.append(mark)
                    break
                else:
                    print("Marks must be between 0 and 100.")

            except ValueError:
                print("Please enter a valid number.")

    student_data.append([subject] + marks)

print("\n===== PERFORMANCE REPORT =====")

for record in student_data:
    subject = record[0]
    exam1 = record[1]
    exam2 = record[2]
    exam3 = record[3]

    average = (exam1 + exam2 + exam3) / 3

    if exam3 > exam1:
        trend = "Improved"
    elif exam3 < exam1:
        trend = "Declined"
    else:
```

```
        trend = "Declined"
    else:
        trend = "Unchanged"

    if average < 40 or exam3 < 40:
        status = "Needs Improvement"
    else:
        status = "Satisfactory"

    print("\nSubject:", subject)
    print("Exam 1:", exam1)
    print("Exam 2:", exam2)
    print("Exam 3:", exam3)
    print("Average:", round(average, 2))
    print("Performance Trend:", trend)
    print("Status:", status)

print("\nAnalysis completed.")
```

10. Conclusion

The Student Performance Trend Analysis program demonstrates how Python can be used to analyze examination results. It uses nested lists, loops, conditional statements, arithmetic operations, and input validation to calculate averages, compare performance trends, and identify subjects that need improvement.